

STA510 – Mandatory assignment 2

Report (theory answers) – autumn 2026

Name: [fill in your name] Student number: [fill in]

This report answers the theory questions of mandatory assignment 2 and comments on the results of the [R] subproblems. The accompanying R-code (mandatory2.R) implements all subproblems marked [R]. All numbers below come from running that script (base R, set.seed(510)), and Figures 1–6 are the plots it produces (also collected in output/mandatory2_plots.pdf).

Problem 1

a) Since $U \sim U[0, 2]$ has density $1/2$, $I = 2 \mathbb{E}[f(U)]$, so $\hat{I}_{CMC} = \frac{2}{n} \sum_{i=1}^n f(U_i)$. With $n = 10000$: $\hat{I}_{CMC} = 0.7899$. (Numerical reference from `integrate()`: $I = 0.79547$.) Figure 1 shows f on $[0, 2]$.

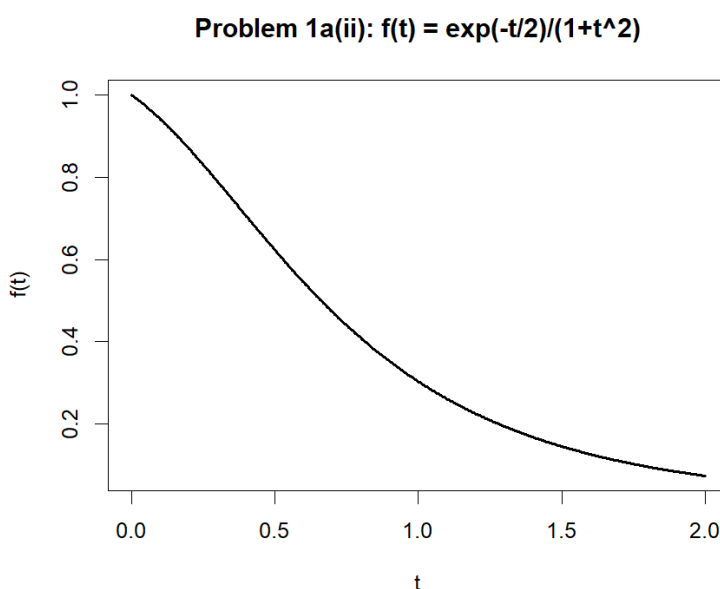


Figure 1: $f(t) = e^{-t/2}/(1+t^2)$ on $[0, 2]$ (Problem 1a(ii)).

b) Yes, f is monotone (decreasing) on $[0, 2]$. By the product rule

$$f'(t) = e^{-t/2} \left[-\frac{1}{2(1+t^2)} - \frac{2t}{(1+t^2)^2} \right] = -\frac{e^{-t/2} (t^2 + 4t + 1)}{2(1+t^2)^2}.$$

For $t \in [0, 2]$ we have $e^{-t/2} > 0$, $(1+t^2)^2 > 0$ and $t^2 + 4t + 1 \geq 1 > 0$, so $f'(t) < 0$.

Antithetic estimator: if $U \sim U[0, 2]$ then $2 - U \sim U[0, 2]$ as well, so $f(U)$ and $f(2 - U)$ both have mean $I/2$. Generate $U_1, \dots, U_{n/2}$ and use

$$\hat{I}_{AT} = \frac{2}{n/2} \sum_{i=1}^{n/2} \frac{f(U_i) + f(2 - U_i)}{2}.$$

This is unbiased and uses the same number (n) of function evaluations as the crude estimator. With $Y_1 = f(U)$, $Y_2 = f(2 - U)$, $\text{Var}\left(\frac{Y_1 + Y_2}{2}\right) = \frac{\sigma^2}{2}(1 + \rho)$ where $\rho = \text{Corr}(Y_1, Y_2)$. Because f is monotone, $f(U)$ and $f(2 - U)$ are monotone functions of U in opposite directions: a large value of one

goes with a small value of the other, so $\rho < 0$. The variance is therefore smaller than with n independent draws ($\rho = 0$). That is why the approach is reasonable here.

c) With $n/2 = 5000$ uniforms: $\hat{I}_{AT} = 0.7971$.

d) See Figure 2. Both f and g are positive and decreasing on $[0, 2]$, so g has roughly the same shape as f . g puts the most mass where f is largest (near $t = 0$), which makes it a reasonable importance function. Also, $g > 0$ wherever $f > 0$, and the ratio

$$\frac{f(t)}{g(t)} = \frac{2(1 - e^{-1})}{1 + t^2} \in [0.253, 1.264]$$

is bounded, so the importance weights cannot blow up. It is not perfect: f decays faster than g because of the extra factor $1/(1 + t^2)$, so g puts too much mass near $t = 2$ and the ratio is not constant. Expect a variance reduction, but not an extreme one.

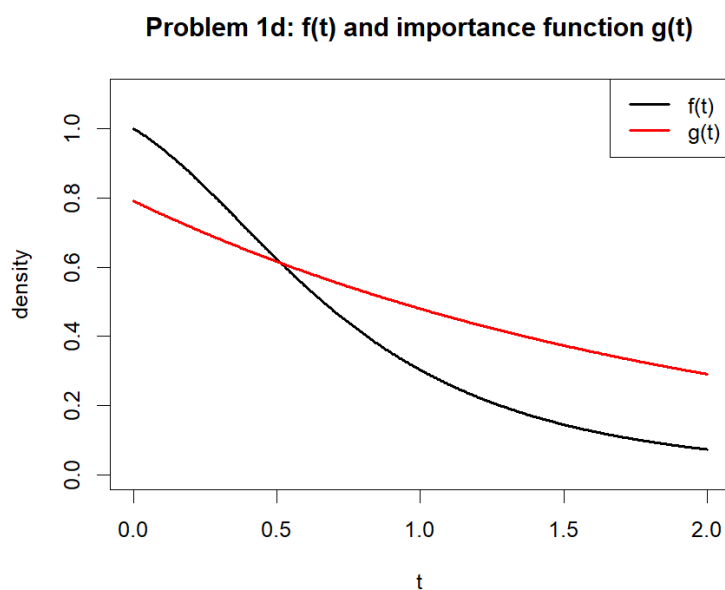


Figure 2: $f(t)$ (black) and the importance function $g(t)$ (red) on $[0, 2]$ (Problem 1d).

e) For $0 \leq t \leq 2$,

$$G(t) = \int_0^t \frac{e^{-s/2}}{2(1 - e^{-1})} ds = \frac{[-2e^{-s/2}]_0^t}{2(1 - e^{-1})} = \frac{1 - e^{-t/2}}{1 - e^{-1}}.$$

(Check: $G(0) = 0$, $G(2) = 1$.) Solving $U = G(t)$ for t :

$$1 - e^{-t/2} = U(1 - e^{-1}) \Rightarrow e^{-t/2} = 1 - U(1 - e^{-1}) \Rightarrow t = -2 \log(1 - U(1 - e^{-1})).$$

So $G^{-1}(U) = -2 \log(1 - U(1 - e^{-1}))$, and by the inverse transform method $G^{-1}(U) \sim g$ when $U \sim U(0, 1)$.

f) With $T_i = G^{-1}(U_i)$, $\hat{I}_{IS} = \frac{1}{n} \sum_{i=1}^n f(T_i)/g(T_i)$. With $n = 10000$: $\hat{I}_{IS} = 0.7977$.

g) 1000 replications of each estimator (each based on $n = 10000$ function evaluations):

Estimator	Mean	SD	Variance reduction vs CMC
CMC	0.795487	0.005666	—
Antithetic	0.795568	0.002097	86.3 %
Importance sampling	0.795327	0.003204	68.0 %

All three means agree with $I = 0.79547$ to about 4 decimals, consistent with all three estimators being unbiased. Both methods reduce the variance a lot. The antithetic estimator does best (variance about 7 times smaller than crude MC): f is monotone and fairly smooth on $[0, 2]$, so $f(U)$ and $f(2 - U)$ are strongly negatively correlated. Importance sampling cuts the variance about 3 times. It helps because g follows the decreasing shape of f , but f/g still varies by a factor of about 5 over $[0, 2]$ (see d), so some variance remains.

Problem 2

a) For a Weibull(k, λ), $E(T) = \lambda \Gamma(1 + 1/k)$ and $\text{Var}(T) = \lambda^2 [\Gamma(1 + 2/k) - \Gamma(1 + 1/k)^2]$. In the tables parametrisation, $\alpha = 1/\lambda^k = 1/36$ and $\beta = k = 2$, so $\alpha^{-1/\beta} = 6$, which gives the same result. With $k = 2$, $\lambda = 6$ and $\Gamma(3/2) = \sqrt{\pi}/2$, $\Gamma(2) = 1$:

$$E(T) = 6 \cdot \frac{\sqrt{\pi}}{2} = 3\sqrt{\pi} \approx 5.317 \text{ months}, \quad \text{SD}(T) = 6\sqrt{1 - \pi/4} \approx 2.780 \text{ months}.$$

b) Simulate inter-failure times $T_i \sim \text{Weibull}(2, 6)$ and add them up until the sum passes 60 months. See Figure 3. In this realisation **9 failures** occurred in $[0, 60]$.

Problem 2b: One realisation of the renewal process

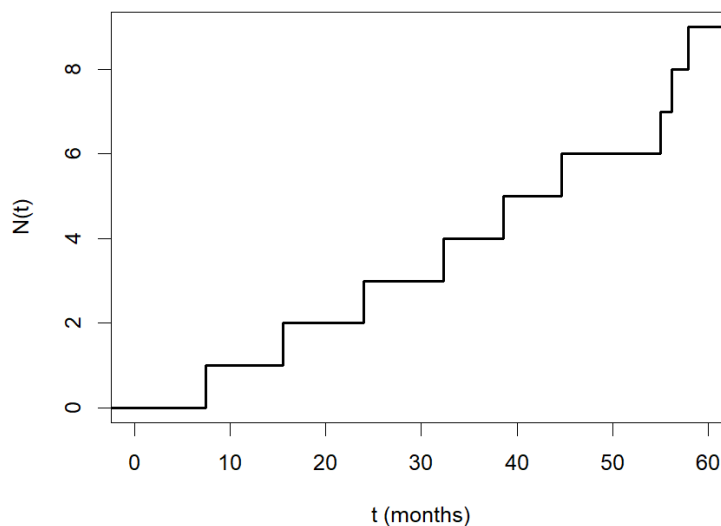


Figure 3: One realisation of the Weibull(2, 6) renewal process $N(t)$ on $[0, 60]$ months, with 9 failures (Problem 2b).

c) Based on $B = 5000$ realisations:

- i) $\hat{E}[N(60)] = 10.94$. For comparison, $t/E(T) = 11.28$ (elementary renewal theorem). The simulated value is a little lower. It matches the second-order approximation

$$m(t) \approx \frac{t}{\mu} + \frac{\sigma^2 - \mu^2}{2\mu^2} = 11.28 - 0.36 = 10.92,$$

which is below t/μ here because $\sigma < \mu$.

- ii) $\widehat{P}(N(60) \geq 15) = 0.030$.

d) Integrating the intensity:

$$\Lambda(t) = \int_0^t \left(3 + 2 \cos \frac{\pi u}{6}\right) du = \left[3u + 2 \cdot \frac{6}{\pi} \sin \frac{\pi u}{6}\right]_0^t = 3t + \frac{12}{\pi} \sin \left(\frac{\pi t}{6}\right).$$

Expected number of failures over 3 years: $\Lambda(36) = 108 + \frac{12}{\pi} \sin(6\pi) = 108$. For an NHPP, $N(36) \sim \text{Poisson}(\Lambda(36))$, i.e. $N(36) \sim \text{Poisson}(108)$.

e) See Figure 4. $\lambda(t)$ has period 12 months. It is highest, $\lambda = 5$, at $t = 0, 12, 24, 36$, i.e. around the turn of the year (December/January, winter). It is lowest, $\lambda = 1$, at $t = 6, 18, 30$, i.e. around the start of July (June/July, summer).

Thinning: $\lambda(t) \leq \lambda_{\max} = 5$. Simulate a homogeneous Poisson process with rate 5 on $[0, 36]$: $N \sim \text{Poisson}(5 \cdot 36)$ points, uniformly placed. Keep each point t_j with probability $\lambda(t_j)/5$. The kept points form an NHPP with intensity $\lambda(t)$. Figure 5 shows one realisation with 133 failures, together with $\Lambda(t)$. Failures cluster in the winter months, and $N(t)$ is nearly flat in the summers. This realisation happens to lie above its expectation ($108 \pm \sqrt{108} \approx 108 \pm 10.4$).

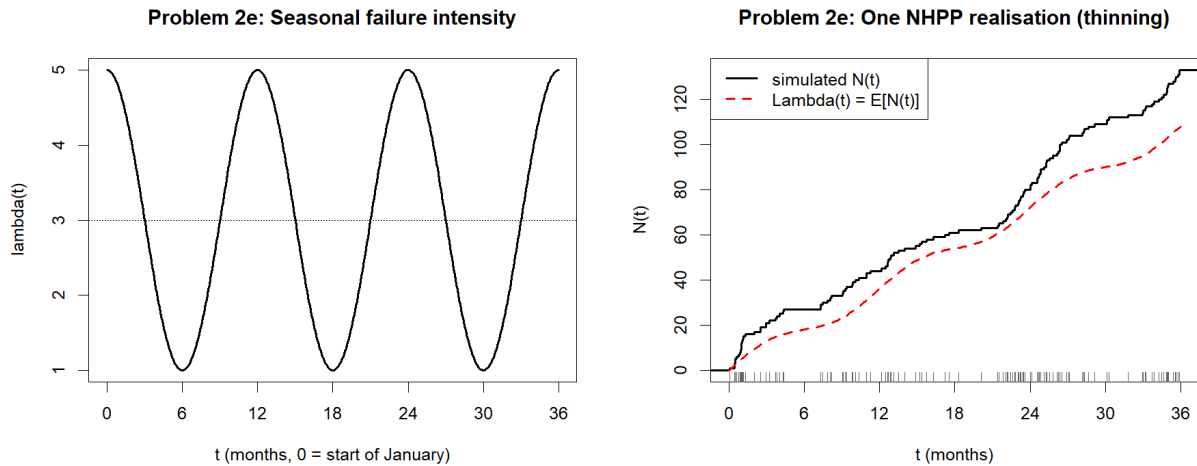


Figure 4 (left): seasonal intensity $\lambda(t) = 3 + 2 \cos(\pi t/6)$ on $[0, 36]$, with the mean level 3 dotted. Figure 5 (right): one NHPP realisation on $[0, 36]$ simulated by thinning (133 failures), with $\Lambda(t) = E[N(t)]$ dashed and the failure times as a rug (Problem 2e).

f) Based on $B = 5000$ realisations:

- i) $\widehat{E}[N(36)] = 108.08$, close to the theoretical $\Lambda(36) = 108$.
- ii) $\widehat{P}(N(36) > 120) = 0.119$. The exact $\text{Poisson}(108)$ value is $1 - F(120) = 0.116$.
- iii) $\widehat{E}[N(3)] = 12.89$. Theory: $\Lambda(3) = 9 + 12/\pi \approx 12.82$. This is well above $3 \cdot 3 = 9$ because the first quarter is the high-failure season.

Problem 3

$n = 40$, $\bar{x} = 7.845$, sample median = 7.40, $s = 3.085$. $B = 2000$ resamples of size n drawn with replacement. Histograms of the bootstrap statistics are in Figure 6.

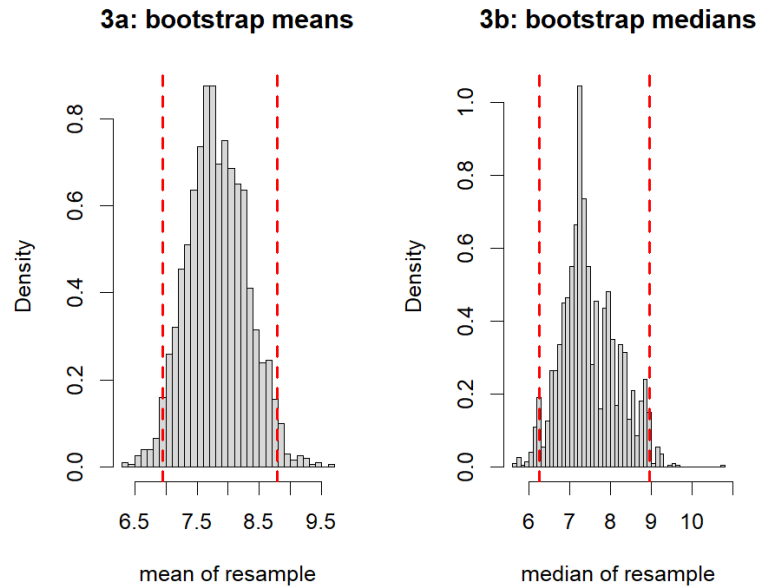


Figure 6: Bootstrap distributions ($B = 2000$) of the sample mean (left) and the sample median (right); the dashed lines are the 95% percentile limits (Problem 3).

a) Mean: bootstrap SD of $\bar{X} = 0.479$, close to $s/\sqrt{n} = 0.488$. 95% percentile bootstrap CI for μ : **(6.947, 8.780)**. The CLT interval from M1 2d was (5.414, 10.276), roughly 2.7 times wider. M1 used the exponential model, where $\text{SD}(X) = \beta$, so the standard error was $\hat{\beta}/\sqrt{n} = 1.24$. The data are much less spread out than an exponential model implies ($s = 3.09$ vs $\hat{\beta} = 7.85$; M1 3d also found a gamma with shape around 6 fits much better). So the model-based interval is too wide. The bootstrap interval assumes no distribution and uses the actual spread in the data. It is close to the normal interval $\bar{x} \pm 1.96 s/\sqrt{n} = (6.89, 8.80)$.

b) Median: bootstrap SD of the sample median = 0.695. 95% percentile bootstrap CI for the population median: **(6.250, 8.950)**. This interval is wider than the one for the mean (length 2.70 vs 1.83) and is slightly asymmetric around 7.40. For roughly symmetric, light-tailed data like these, the median is a less efficient estimator than the mean, so it varies more. The bootstrap distribution of the median is also lumpy and discrete (Figure 6): with $n = 40$ the median of a resample is the average of its 20th and 21st order statistics, i.e. the average of two of the original data values, so only a limited set of values is possible. The percentile interval for the median is therefore cruder, and less reliable with $n = 40$, than the one for the mean. Also note the two intervals target different parameters (population mean vs median), which differ because the data are slightly right-skewed.

References

- STA510 lecture notes (Monte Carlo integration and variance reduction; Poisson and renewal processes; bootstrap) and the tables/formulas sheet.
- Rizzo, M. L., *Statistical Computing with R*, ch. 6 (antithetic variables, importance sampling).